1. (10 marks)

State whether each of the following statements is true or false.

[Justifications are not required.]

(a) If X is a set, then any collection of subsets of X forms the basis for a topology on X.

(b) Suppose A is an arbitrary indexing set and for each $\alpha \in A$, X_α is a set. Let

$$X = \prod_{\alpha \in A} X_\alpha,$$

let σ be the product topology on X, and let τ be the box topology on X. Then the function

$$f : (X, \sigma) \to (X, \tau),$$

where f is the identity function on X, is continuous.

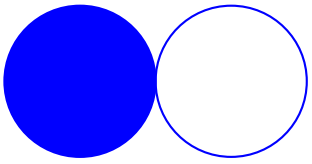
(c) If X is a topological space and A is an open subset of X, then $\text{int}(\text{cl}(A)) = A$.

(d) Let X be a connected topological space. Then the cone space on X is connected.

(e) Let A be an indexing set and let $A = B \cup C$ such that $B \neq \emptyset \neq C$. Suppose that for each $\alpha \in A$, X_α is a topological space, for each $\alpha \in B$, X_α is separable, and for each $\alpha \in C$, X_α is first countable. Then $\prod_{\alpha \in A} X_\alpha$ is first countable.

(f) All metric spaces are second countable.

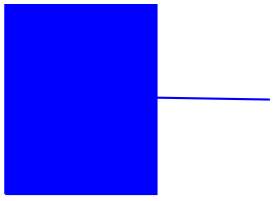
(g) The fundamental group of the structure shown in the diagram below is $\mathbb{Z}$.



(h) Any two loops on a cylinder are homotopic.

(i) Suppose X, Y, Z are topological spaces. If $p : X \to Y$ is a covering projection and $q : Y \to Z$ is a covering projection, then $q \circ p : X \to Z$ is a covering projection.

(j) The space in the diagram below, a solid rectangle with a line segment attached, has a Δ-complex structure.



2. (15 marks)

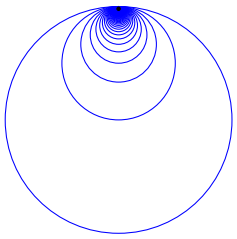
(a) Let X and Y be topological spaces, and let $f : X \to Y$ be a function. Prove that the following two statements are equivalent.

i. For each $A \subseteq Y$, $\overline{f^{-1}(A)} \subseteq f^{-1}(\overline{A})$.

ii. For each $A \subseteq Y$, $f^{-1}(A^\circ) \subseteq (f^{-1}(A))^\circ$.

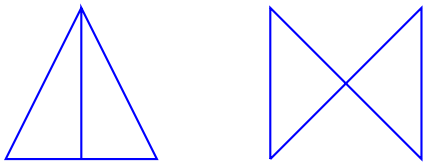
[Hint: Note that $\overline{A} = X \setminus A^\circ$.]

(b) Is the Hawaiian earring continuum, infinitely many circles that meet at a point as shown in the diagram below, locally path-connected? Give a brief justification for your answer.



The Hawaiian earring

(c) Are the following two finite graphs homeomorphic? Give a brief justification for your answer.



3. (15 marks)

   (a) Show that every compact Hausdorff space is regular.

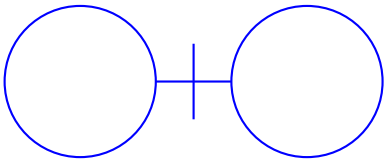
   (b) Let $X = \mathbb{N} \times [0, 1]$ endowed with the subspace topology inherited from $\mathbb{R}^2$. Let X^* be the one-point compactification of X. Is X^* is locally connected? Justify your answer

   (c) Let $\mathbb{P}$ be the set of irrational numbers in $\mathbb{R}$ endowed with the subspace topology. Let $X = \mathbb{P} \cup \{\infty\}$ and let τ be the topology on X that is the one-point compactification of $\mathbb{P}$. Is (X, τ) first countable? Justify your answer.

4. (15 marks)

   (a) Let X be the figure as shown in the diagram below. Give an example of a covering space E and a covering projection $p : E \to X$, such that it is not the case that X itself is evenly covered.

   [You do not need to define E and p formally, a diagram is adequate.]



   (b) Find the simplicial homology groups for the structure shown in the diagram below. Show all working.

